Magnetosphere and changing fields

antocciphysics | TEACHER KEY | CPA + MATCHING SEI

Learning target: Use evidence to distinguish a magnetic-field model from an unsupported space-weather claim.

Accept equivalent correct methods and clearly labeled models. Language supports do not change the physics demand. Supplied data are examples; actual lab measurements and teacher checkpoints must be recorded from the lesson.

01. Warm-up: bring back an earlier idea *Review: Unit 7, Day 13.*
New field is 1/3 of the original under the long-wire model.

02. Worked example: explain the model
Change field magnitude/orientation or loop area; relevant quantity is flux change through the loop.

03. Read and interpret the model
Changing flux through coil; emf around loop.

04. Apply the physics
4 V in the proportional model.

05. Explain your reasoning
A credible physical mechanism, measured/predicted field change and timing, source attribution, conductor geometry and current estimates; proximity alone is insufficient.

06. Review: connect an earlier idea *Review: Unit 7, Day 4.*
6 J of electric potential-energy difference per coulomb.

07. Exit check
No; a flux change is required.

08. STRETCH: test the model
Larger area gives greater emf for the same uniform dB/dt and one turn; real geometry and conductivity matter.

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01. Warm-up: bring back an earlier idea *Review: Unit 7, Day 13.*
$B_f/B_i = 3/5 \approx 0.6$, using $B \propto I/r$.

02. Worked example: explain the model
Change field magnitude/orientation or loop area; relevant quantity is flux change through the loop.

03. Read and interpret the model
Changing flux through coil; emf around loop.

04. Apply the physics
$|\text{emf}| = NA|\Delta B|/\Delta t = (10)(5)(2 \times 10^{-6})/7 \approx 1.43 \times 10^{-5}$ V. No: the result follows from a hypothetical field change and does not establish its cause.

05. Explain your reasoning
A credible physical mechanism, measured/predicted field change and timing, source attribution, conductor geometry and current estimates; proximity alone is insufficient.

06. Review: connect an earlier idea *Review: Unit 7, Day 4.*
32 J and 48 J; total 80 J.

07. Exit check
Real networks have geometry, grounding, resistance, controls and spatially varying fields; a justified larger model and observations are required.

08. STRETCH: test the model
$\text{Emf} = A\Delta B/\Delta t = 0.0004$ V. Real grids involve geometry, ground conductivity, spatial/temporal variation and network response.